

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
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Course Outcome	C03-DFS in Microservice Call Graphs

Case Study Topic: Microservice Call-Graph DFS — Downstream Service Discovery for Distributed Trace-IDs

Work Done Summary:

In this case study, we implemented Depth-First Search (DFS) on a directed microservice call-graph with 9 services and 12 edges, exploring neighbors in alphabetical order starting from the **api** gateway. We determined the entry and post-order finish times for all vertices, identified transitively reachable downstream services, and analyzed the $O(V + E)$ time complexity to show how distributed tracing tools like Jaeger and Zipkin diagnose system latency.

Implementation Details:

1. Created a directed graph structure using an adjacency list to map microservices and their dependency connections.
2. Ingested 9 distinct service vertices and linked them using 12 directional dependencies.
3. Sorted all neighbor dependency lists alphabetically to guarantee a deterministic traversal path.
4. Implemented a recursive Depth-First Search (DFS) function tracking a global logic clock.
5. Recorded exact vertex entry order and post-order finish times during graph exploration.
6. Identified all transitively reachable downstream services originating directly from the **api** gateway.

Result: Screenshots/output:

```
Visited Nodes (Entry Order):
[api, auth, analytics, cart, inventory, payment, notif, ship, catalog]

Entry Times:
{api=1, auth=2, analytics=3, cart=6, inventory=7, payment=9, notif=10,
 ship=12, catalog=16}

Finish Times (Post-Order):
{analytics=4, auth=5, inventory=8, notif=11, ship=13, payment=14, cart
 =15, catalog=17, api=18}

Downstream Services from API:
[api, auth, analytics, cart, inventory, payment, notif, ship, catalog]

=== Code Execution Successful ===
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GitHub Link: https://github.com/Varnika1024/DSA_CASE-STUDY/blob/main/C03.java